

Zero-Shot Evaluation of gemini-3.5-flash on AfriXNLI

A Case Study within the AfroBench-Lite Benchmark

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Abstract

This report presents a zero-shot evaluation of gemini-3.5-flash on AfriXNLI within AfroBench-Lite, across all 14 designated languages and five prompt templates. We report per-language, per-prompt F1/accuracy and the best-performing prompt per language (T1–T5), achieving a macro-average of 77.64 F1, with T3 generalizing best as a single fixed prompt across languages (77.12 F1). Two additional findings are highlighted: a near-chance failure mode on Lingala across all five prompts, and a content-safety filtering artifact on T5 that measurably depresses its aggregate score relative to the other four prompts.

1 Background

This report concerns AfriXNLI, the Natural Language Inference (NLI) task within AfroBench-Lite: given a premise and a hypothesis, the model classifies their relationship as entailment, neutral, or contradiction. As with other AfroBench-Lite tasks, AfriXNLI is evaluated across five prompt templates per language, with the best-performing prompt reported per language following the paper’s own convention (Table 12, using T1–T5 notation).

2 Methodology

The model and task were configured as below:

- Model evaluated: gemini-3.5-flash, accessed via the Gemini API.
- Task: AfriXNLI, using afrobench lite/afrixnli.yaml.
- Languages (14): eng, amh, hau, ibo, kin, lin, lug, orm, sna, sot, swa, xho, yor, zul.
- Shot setting: Zero-shot.
- Prompts: All five templates (T1–T5) run independently; best prompt reported per language.
- Metrics: F1 (macro) and accuracy, via metrics.py’s verbalizer-based scoring.

3 Results

3.1 Best-Performing Prompt per Language

Table 1 reports the best-performing prompt (by F1) for each of the 14 languages.

Table 1: Best-performing prompt per language for gemini-3.5-flash on AfriXNLI (zero-shot).

Language	Best Prompt	F1	Accuracy
Amharic (amh)	T3	82.83	82.83
English (eng)	T4	90.17	90.17
Hausa (hau)	T4	80.50	80.50

Igbo (ibo) T1 82.33 82.33
 Kinyarwanda (kin) T4 74.33 74.33
 Lingala (lin) T5 33.33 33.33
 Luganda (lug) T3 78.67 78.67
 Oromo (orm) T1 81.67 81.67
 chiShona (sna) T3 82.83 82.83
 Sesotho (sot) T1 78.67 78.67
 Kiswahili (swa) T3 80.50 80.50
 isiXhosa (xho) T3 84.33 84.33
 Yoruba (yor) T3 75.50 75.50
 isiZulu (zul) T3 81.33 81.33
 Macro-average (14 languages) – 77.64 77.64

3.2 Per-Prompt Macro-Average

Table 2 reports the macro-average across all 14 languages for each prompt held fixed. T3 is the single best fixed prompt overall, and is also individually best for 6 of the 14 languages – more than any other template.

Table 2: Macro-average F1/accuracy across all 14 languages, per fixed prompt.

Prompt F1 Accuracy
 T1 76.14 76.14
 T2 70.62 70.62
 T3 77.12 77.12
 T4 76.92 76.92
 T5 65.17 65.17

4 Discussion

4.1 Lingala: A Near-Chance Failure Mode

Lingala (lin) scores 32–33% F1 across all five prompts – indistinguishable from random guessing among the three NLI labels, and flat where every other language shows prompt-to-prompt variation. This points to a genuine competence gap rather than prompt sensitivity: the model likely cannot reliably process Lingala, consistent with its low representation in typical LLM pretraining data.

4.2 T5 and Content-Safety Filtering

T5 is the weakest prompt overall (65.17 F1 vs. 76–77 for T1, T3, T4). A subset of T5 requests triggered Gemini’s safety filter (block reason: PROHIBITED CONTENT), returning no response. This reflects the

model declining to answer certain inputs under this prompt's phrasing, not a classification error – and likely deflates T5's score below its true capability on that content.

5 Conclusion

gemini-3.5-flash achieves 77.64 macro F1 across the 14 AfroBench-Lite languages on AfriXNLI (zero-shot, best prompt per language), with T3 the strongest single fixed prompt (77.12 F1). Two factors should be considered when interpreting these results: Lingala performance is at chance regardless of prompt, indicating a language-specific model limitation rather than a prompting issue; and T5's score is partly suppressed by safety-filter refusals rather than genuine misclassification.